Jacob Torry

jacobmtorry@gmail.com | (309)-445-2781 | linkedin.com/in/jacobtorry | jacobmtorry.github.io

Education

University of Illinois Urbana-Champaign, BS in Computer Engineering — Aug 2022 – May 2027

- GPA: 3.32/4.0
- Minor in Semiconductor Engineering
- **Relevant Coursework:** Computer Organization & Design, Digital Systems Lab, Computer Architecture, Parallel Programming (CUDA), Semiconductor Electronics, Signal Processing, AI

Work Experience

Intern, SCADAware – Normal, IL — Jun 2025 – Aug 2025

- Delivered industrial automation solutions including robotic systems, motor control panels, and PLC-based control systems.
- Produced electrical schematics, panel layouts, and wiring diagrams in AutoCAD for industrial control hardware.
- Collaborated with multidisciplinary teams to translate system requirements into deployable control solutions.

Projects

Out-of-Order RISC-V Processor

- Designed and implemented an out-of-order RISC-V processor in SystemVerilog, including register renaming, reservation stations, reorder buffer (ROB), load-store queue, and speculative execution.
- Achieved the highest clock frequency in the class upon submission through aggressive pipeline optimization and critical-path reduction.
- Verified functional correctness using simulation testbenches and ISA-level validation against the Spike reference model.

FPGA Pacman Remake

- Implemented a hardware-accelerated Pacman game on FPGA using SystemVerilog, including custom video controller, sprite engine, and memory-mapped peripherals.
- Designed AXI-based communication between hardware modules and embedded processor; implemented USB keyboard input via SPI.
- Optimized BRAM usage and memory layout for tilemap, sprite, and font storage.

GPU Parallel Programming

- Developed CUDA kernels for matrix multiplication, convolution, and parallel reductions using shared memory and tiled algorithms.
- Optimized GPU performance by minimizing global memory traffic and exploiting memory coalescing and thread-level parallelism.
- Analyzed performance using roofline modeling and profiling tools.

Illinix- Basic Unix Operating system

- Developed and debugged a Unix-like operating system in RISC-V and C as part of a 3-person team.
- Implemented a file system enabling the Virtual I/O block device to open, close, read, and write files.
- Created test cases and used GDB to debug and validate memory operations, collaborating with teammates to ensure reliability.

Technologies

HDL / Hardware: SystemVerilog, Verilog, RISC-V ISA, FPGA Design, Microarchitecture
Programming: C, C++, CUDA, Python
Tools: Vivado, Vitis, Synopsys, Git, Linux, GDB, ModelSim
Platforms: FPGA (RealDigital Urbana), Raspberry Pi
Semiconductor: SEM imaging, Crosslight APSYS TCAD